THE CLASSIFYING STACK OF A GROUP ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1675-1693 of sources/stacks-examples-stacks/pinned-original-source.tex; proof span is lines 1695-1725, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_examples-stacks.json (pb=1463, stmt 1675-1693, proof 1695-1725, halfA).

Editorial note. stacks-examples-stacks wave-1 (exs1): fresh adjudication accept (2026-10-14); pool per Chief 09:4x authorization (idle-line throughput over kun's backup note, SMC precedent); examples-stacks.tex 27 labels/20 with proof judged 9 accept / 11 reject; claim_key v2 stacks:examples-stacks.tex:<label>. Statement carries a slogan environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04UT: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1358–1369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04WM: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1528–1536. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04UV: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1409–1426. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041W: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1725–1725. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041T: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1725–1725. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041Q: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1725–1725. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04TY: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1725–1725. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The classifying stack of a group scheme or group algebraic space. Let S be a scheme. Let B be an algebraic space over S. Let G be a group algebraic space over B. Then the stacks in groupoids*

$$[B/G], \quad [[B/G]], \quad G\text{-}Torsors, \quad \mathcal{G}/\mathcal{B}\text{-}Torsors$$

are all canonically equivalent. If $G \to B$ is flat and locally of finite presentation, then these are also equivalent to G-Principal.

Proof. The equivalence $G\text{-}Torsors \to \mathcal{G}/\mathcal{B}\text{-}Torsors$ is given in Lemma 04UT. The equivalence $[B/G] \to [[B/G]]$ is given in Proposition 04WM. Unwinding the definition of $[[B/G]]$ given in Section 04UV we see that $[[B//G]] = G\text{-}Torsors$.

Finally, assume $G \to B$ is flat and locally of finite presentation. To show that the natural functor $G\text{-}Torsors \to G\text{-}Principal$ is an equivalence it suffices to show that for a scheme U over B a principal homogeneous G_U-space $X \to U$ is fppf locally trivial. By our definition of principal homogeneous spaces (Groupoids in Spaces, Definition 04TY) there exists an fpqc covering $\{U_i \to U\}$ such that $U_i \times_U X \cong G \times_B U_i$ as algebraic spaces over U_i. This implies that $X \to U$ is surjective, flat, and locally of finite presentation, see Descent on Spaces, Lemmas 041Q, 041W, and 041T. Choose a scheme W and a surjective étale morphism $W \to X$. Then it follows from what we just said that $\{W \to U\}$ is an fppf covering such that $X_W \to W$ has a section. Hence X is an fppf G_U-torsor. □